

Further Statistics 1

Notes · Edexcel A Level Further Mathematics — Further Statistics 1 · Chapter 8: Quality of Tests

Chapter 8: Quality of Tests

Why this chapter matters

Every hypothesis test you have carried out so far ends with a binary decision: reject H_0 or don't. But that decision can be wrong in two distinct ways, and a good statistician needs to think critically about *how* wrong, and how likely each kind of error is. This final chapter steps back from performing tests to evaluating them: understanding Type I and Type II errors, and the closely related idea of the power of a test. This is exactly the kind of conceptual, less-calculation-heavy material that exam papers love to test with “explain” and “comment” questions.

Concept: Type I and Type II errors

When carrying out a hypothesis test, two types of error are possible:

- **Type I error** occurs when H_0 is rejected when it is actually true.
- **Type II error** occurs when H_0 is not rejected when it is actually false (H_1 is true).

These are the only two ways a test can go wrong, and they trade off against each other: making a test less likely to commit a Type I error (e.g. by lowering the significance level) generally makes it more likely to commit a Type II error, and vice versa.

A memory hook: think of a criminal trial, where H_0 is “the defendant is innocent.” A **Type I error** is convicting an innocent person — rejecting a true H_0 . A **Type II error** is acquitting a guilty person — failing to reject a false H_0 . Courts set a high bar (“beyond reasonable doubt”) precisely because they treat a Type I error as the more serious mistake to guard against, even though this makes Type II errors (letting a guilty person go free) more likely — exactly the trade-off described above, just with real-world stakes attached instead of a significance level.

Key vocabulary

Term	Meaning
Type I error	Rejecting a true H_0
Type II error	Failing to reject a false H_0
Significance level, α	$P(\text{Type I error}) = P(\text{rejecting } H_0 \mid H_0 \text{ true})$
$P(\text{Type II error}), \beta$	$P(\text{not rejecting } H_0 \mid H_1 \text{ true})$, which depends on the TRUE value of the parameter under H_1
Power of a test	$1 - \beta = P(\text{correctly rejecting } H_0 \mid H_1 \text{ true})$

Key formula: calculating P(Type I error) and P(Type II error)

- $P(\text{Type I error}) = P(\text{test statistic falls in the critical region} \mid H_0 \text{ is true})$. For a discrete test statistic, this is exactly the actual significance level found when the critical region was constructed in Chapter 4 (which is why it usually isn't exactly equal to the nominal 5% or 1%).
- $P(\text{Type II error})$ requires you to specify a particular true value of the parameter under H_1 , then calculate $P(\text{test statistic falls OUTSIDE the critical region, calculated using this new true value})$.

Worked example 1: calculating P(Type I error)
EXAMPLE 8.1

Question. $X \sim B(20, p)$ is used to test $H_0: p = 0.3$ against $H_1: p > 0.3$ at the 5% significance level. The critical region was found to be $X \geq 10$. Find the actual probability of a Type I error.

Step 1. $P(\text{Type I error}) = P(X \geq 10 \mid p = 0.3)$, i.e. using the distribution under H_0 . **(M1)**

Step 2. From $B(20, 0.3)$ tables, $P(X \leq 9) = 0.9520$, so $P(X \geq 10) = 1 - 0.9520 = 0.0480$. **(M1A1)**

Answer: $P(\text{Type I error}) = 0.0480$ (this is the *actual* significance level of the test, slightly below the nominal 5% because X is discrete).

EXAM TIP

Make the point explicitly that $P(\text{Type I error})$ is calculated using H_0 's parameter value, because a Type I error is by definition about wrongly rejecting a TRUE H_0 — so we must work under the assumption that H_0 is true when computing it. This sounds obvious once stated but is a common point of confusion, especially once Type II errors (calculated under H_1 's value) are introduced in the same topic.

Worked example 2: calculating P(Type II error) and power
EXAMPLE 8.2

Question. Using the same test as Example 1 ($H_0: p = 0.3$ vs $H_1: p > 0.3$, $X \sim B(20, p)$, critical region $X \geq 10$), suppose the true value of p is actually 0.5. Find the probability of a Type II error, and the power of the test at $p = 0.5$.

Step 1. A Type II error means failing to reject H_0 , i.e. X falls outside the critical region, i.e. $X \leq 9$ — but now calculated using the TRUE value $p = 0.5$, not the H_0 value. **(M1)**

Step 2. $P(\text{Type II error}) = P(X \leq 9 \mid p = 0.5)$. From $B(20, 0.5)$ tables, $P(X \leq 9) = 0.4119$. **(M1A1)**

Step 3. Power = $1 - P(\text{Type II error}) = 1 - 0.4119 = 0.5881$. **(M1A1)**

Answer: $P(\text{Type II error}) = 0.4119$, Power = 0.5881. This means that if the true proportion really has risen to 0.5, this test only has a 58.8% chance of correctly detecting that rise from a single sample of size 20.

EXAM TIP

This is a good moment to think about WHY power matters in real terms: a low power (as here, under 60%) means the test frequently fails to detect a genuine effect, which is exactly as serious a problem as false positives, but often gets less attention. Try recalculating power at $p = 0.4$ vs p

= 0.7 to see power increasing as the true parameter moves further from H_0 — the further apart H_0 and reality are, the easier a test finds it to “notice.”

Concept: factors affecting power

The power of a test increases (Type II error decreases) when:

1. The sample size n increases (more data makes it easier to detect a real effect) — this is the same idea as the Central Limit Theorem's σ^2/n shrinking as n grows: a larger sample pulls the test statistic's distribution tighter around the true parameter, so H_0 's and H_1 's distributions overlap less and are easier to tell apart.
2. The true parameter value moves further away from the H_0 value (a bigger effect is easier to detect).
3. The significance level α is increased (a larger critical region makes rejection more likely, but at the cost of a higher Type I error rate — this is the fundamental trade-off).

Worked example 3: comparing power for two different sample sizes

EXAMPLE 8.3

Question. A test of $H_0: \lambda = 4$ against $H_1: \lambda > 4$ for a Poisson variable X uses the critical region $X \geq 8$ when one observation window is used (i.e. $X \sim \text{Po}(4)$ under H_0). Now suppose the observation window is doubled, so under H_0 , $Y \sim \text{Po}(8)$, with critical region $Y \geq 14$ (found by the same method as Chapter 4, chosen to keep a similar significance level). If the true rate is actually $\lambda = 6$ per original window (so 12 over the doubled window), compare the power of the two tests.

Step 1. For the original test: power = $P(X \geq 8 \mid \lambda = 6)$, using $\text{Po}(6)$. From tables, $P(X \leq 7) = 0.7440$, so power = $1 - 0.7440 = 0.2560$. **(M1A1)**

Step 2. For the doubled-window test: power = $P(Y \geq 14 \mid \text{true mean} = 12)$, using $\text{Po}(12)$. From tables, $P(Y \leq 13) = 0.6815$, so power = $1 - 0.6815 = 0.3185$. **(M1)**

Step 3. Since $0.3185 > 0.2560$, the doubled-window test has the higher power. This illustrates the general principle: collecting more data (a longer observation window, or a larger sample) increases the power of a test to detect a genuine shift in the parameter, for a fixed significance level. **(A1)**

EXAM TIP

Numerical examples comparing power across two different sample sizes can get algebraically heavy, so it is often more valuable to reason about the DIRECTION of the effect (larger $n \Rightarrow$ higher power, for fixed α) using the intuition of “more data reduces random noise around the true value,” and reserve exact calculation for cases with clean numbers. Exam mark schemes frequently reward this reasoning even when a full numerical comparison is not requested.

Chapter summary

Idea	Key point
------	-----------

Type I error	Reject H_0 when H_0 is true; probability = the actual significance level, calculated using the H_0 parameter
Type II error	Fail to reject H_0 when H_1 is true; probability calculated using a SPECIFIED true parameter value under H_1
Power	$1 - P(\text{Type II error})$; the probability of correctly detecting a genuine effect
Trade-off	Lowering α (fewer Type I errors) generally increases $P(\text{Type II error})$, and vice versa
Increasing power	Larger sample size, or true parameter further from the H_0 value, increases power

Common exam mistakes

- Calculating $P(\text{Type I error})$ using the H_1 parameter value instead of the H_0 value (or vice versa for Type II error) — always match the error type to the correct hypothesis's parameter.
- Forgetting that a Type II error calculation requires a SPECIFIC numerical value under H_1 to be given or assumed — unlike $P(\text{Type I error})$, it cannot be calculated from $H_1: p > 0.3$ alone without a specific value such as $p = 0.5$.
- Confusing power ($1 - \beta$) with the significance level (α) — they concern opposite types of error.
- Assuming that a smaller significance level is always “better” without acknowledging the corresponding increase in Type II error risk.
- Giving a vague verbal explanation of Type I/Type II errors without correct reference to which hypothesis is true and which decision was made.

You should now be able to

- Define Type I and Type II errors precisely, in terms of which hypothesis is actually true and which decision is made.
- Calculate $P(\text{Type I error})$ for a given critical region and significance test.
- Calculate $P(\text{Type II error})$ and the power of a test for a specified true parameter value under H_1 .
- Explain how sample size, the significance level, and the true parameter value each affect the power of a test.